Nataly R. Panczyk

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Education

The University of Michigan – Ann Arbor — In Progress
PhD in Nuclear Engineering and Radiological Sciences
Research Focus: AI for Safety-Critical Applications
GPA: 4/4

The University of Illinois Urbana-Champaign — May 2024
BS in Nuclear, Plasma, and Radiological Engineering
Concentration: Power, Safety, and the Environment
Minor: Political Science
GPA: 3.94/4

Work Experience

Guest Researcher – Los Alamos National Laboratory — June 2025 – April 2026
Los Alamos, New Mexico

- Development of an explainable machine learning-based neutron resonance fitting algorithm using a 1D convolutional neural network.

Research Assistant –Multiphase Thermo-fluid Dynamics Laboratory at UIUC — Jan. 2023 – Jan. 2024
Urbana, Illinois

- Analysis of post-critical heat flux (CHF) conditions for downward flow boiling in a vertical square channel
- Responsible for maintenance and operation of the High Heat Flux Test Section (HHFTS)
- Developed data visualization, animation, and analysis code in Python for experimental results from the HHFTS
- Created data visualizations and processing techniques for nuclear opinion survey data in collaboration with The International Institute for Carbon- Free Energy Research hosted at Kyushu University in Fukuoka, Japan
- Mentored by Prof. Caleb Brooks

Intern – MPR Associates — May 2023 – July 2023
Alexandria, Virginia

- Analyzed a reactivity insertion accident for Ultra Safe Nuclear Corporation's Micro Modular Reactor using RELAP-3D and developed the methodology for future analyses
- Developed a maintenance dose tool using Python for a fusion system
- Drafted a white paper on coupling municipal wastewater systems with nuclear microreactors for the Electric Power Research Institute

Research Assistant – Advanced Reactors and Fuel Cycles at UIUC — Sep. 2020 – Dec. 2022
Urbana, Illinois

- Began TRISO fuel analysis with MOOSE for the UIUC microreactor project
- Designed techno-economic analyses for clean hydrogen systems
- Evaluated technological feasibility of fountain-style reactor
- Served as group technical editor for all publications after completing *Writing in the Sciences* from Stanford University via Coursera
- Mentored by Prof. Kathryn Huff and Dr. Madicken Munk

Intern – Pacific Northwest National Laboratory June 2022 – Aug. 2022
Richland, Washington

- Participated in the Science Undergraduate Laboratory Internship (SULI) program through the U.S. Department of Energy (DOE)
- Automated quality assurance techniques for GoldSim model of U.S. spent nuclear fuel management in Python to transition from manual process
- Created default data visualizations to couple with GoldSim model outputs, including an automatic Gantt chart and risk mitigation diagram

Intern – Idaho National Laboratory May 2021 – Aug. 2021
Idaho Falls, Idaho

- Conducted a techno-economic analysis on the feasibility of implementing microreactors into a clean steel manufacturing scheme using Temoa, an open-source energy system optimization tool created by North Carolina State University
- Drafted a nearly 90-page report culminating findings of the clean steel analysis, including appendices considering cybersecurity and licensing implications for microreactors
- Mentored by Dr. Piyush Sabharwall

Research Publications

N. Panczyk, A. Stamatopoulos, J. Svoboda, M. Radaideh, "Learning to Resolve Neutron Resonances with Fully Convolutional Neural Networks" Under review with *Nuclear Science and Engineering*, 2026. 2026

N. Panczyk and M. Radaideh, "When computers call the shots —a systematic literature review exploring the impact of AI in high-stakes decision-making," Accepted for publication in *Artificial Intelligence Review*, 2026. 2026

N. Panczyk, A. Stamatopoulos, J. Svoboda, M. Radaideh, "Automatically Resolving Neutron Resonances with Machine Learning," Accepted for the International Conference on Physics of Reactors, 2026. 2026

N. Panczyk, O. Erdem, M. Radaideh, "Opening the AI Black-Box: Symbolic Regression with Kolmogorov-Arnold Networks for Advanced Energy Applications," Energy and AI, Volume 22, 2025, 100595. 2025

P. Myers, N. Panczyk, S. Chidige, C. Craig, J. Cooper, V. Joynt, M. Radaideh, "pyMAISE: A Python platform for automatic machine learning and accelerated development for nuclear power applications," Progress in Nuclear Energy, Volume 180, 2025, 105568. 2025

O. Erdem, N. Panczyk, M. Radaideh, "Interpretable Machine Learning Regression for Nuclear Applications with Kolmogorov-Arnold Networks (KAN)," Accepted for the American Nuclear Society's M&C Conference, 2025. 2025

A. Xu, N. Panczyk, M. Radaideh, "Towards Explainable AI in Nuclear: Introducing Ad Hoc Model Explainability," Accepted for the American Nuclear Society's M&C Conference, 2025. 2025

N. Panczyk, "Analysis of Post-CHF Conditions for Downward Vertical Boiling in a Square Channel," presented at the American Nuclear Society Student Conference, Knoxville, Tennessee, April 14, 2023. 2023

N. Panczyk, C. Spitz, J. Gastelum, "Quality Assurance and Data Visualization Methods for Spent Nuclear Fuel Modeling," poster presented at the Waste Management Symposia, Phoenix, Arizona, February 26, 2023. 2023

N. Panczyk, N. Ryan, M. Munk "Illinois Energy System Models and the Potential for Proactive Nuclear Activism," poster presented at the American Nuclear Society Student Conference, Champaign, Illinois, April 15, 2022. — 2022

N. Panczyk, S. Dotson, M. Munk, P. Sabharwall, "Preliminary Technoeconomic Analysis for Microreactor Implementation in the Steel Industry," tech. rep., Idaho National Laboratory, August 2021. — 2021

S. Dotson, A. Bachmann, Z. Richter, N. Panczyk, N. Ryan, A. Balla, E. Fanning, K. Huff, M. Munk, "Economic and Carbon Impacts of Potential Illinois Nuclear Plant Closures," tech. rep., University of Illinois Urbana-Champaign, May 2021. — 2021

Invited Talks

Harper Academy Lecture – Ann Arbor, MI — July 17, 2026
- Seminar for Washtenaw County high schoolers on safety-critical AI

Alchemy3 Annual Retreat – Nashville, TN — Feb. 2, 2026
- Seminar on safe use of generative AI in the workplace

ML for Nuclear Engineering Guest Lecture – Ann Arbor, MI — Jan. 2026
- Linear regression methods, including Lasso and Ridge regressions, NERS 590

Teaching Experience

Teaching Assistant: NPRE 455 – UIUC — Jan. 2024 – May 2024
- Held weekly office hours for NPRE 455: Neutron Diffusion and Transport
- Graded problem sets and computer projects

Lead Engineering Learning Assistant (ELA) – UIUC — Aug. 2023 – Jan. 2024
- Served as NPRE lead representative and instructor for Engineering 100 (ENG 100), an orientation class organized by major required for all students in the Grainger College of Engineering
- Redesigned final project curriculum to more accurately reflect skills required for success as an undergraduate engineering student
- Led weekly department meetings for all NPRE ELAs
- Represented NPRE at weekly, college-wide ENG 100 meetings for all department leads
- Taught ENG 100 semiweekly

Engineering Learning Assistant – UIUC — Aug. 2022 – Dec. 2022
- Taught ENG 100 semiweekly
- Held 1-1 mentoring sessions for all students, three times over the course of the semester
- Designed and offered an introductory Python workshop for freshmen
- Graded and provided feedback on weekly assignments

Teaching Assistant: NPRE 247 – UIUC — Jan. 2022 – May 2022
- Held weekly office hours for NPRE 247: Modeling Nuclear Energy Systems, which covers basic nuclear physics and engineering concepts (kinematics, Q-value calculations, number density calculations, etc.)
- Graded all problem sets, reviewed exam material, and graded two of three computer projects

Teaching Assistant: NPRE 100 – UIUC — Aug. 2021 – Dec. 2021
- Held office hours for NPRE 100: Orientation to NPRE, which covers basic content for NPRE undergraduates (intro to Python programming, 4-year class planning, and lab report fundamentals)
- Responsible for grading Python projects (three-part, basic programming project emphasizing nuclear applications of computational methods)

Professional Activities and Community Engagement

Culture Committee Member – University of Michigan Nuclear Engineering Dept. Oct. 2025 – Present

Co-President – Women in Nuclear UIUC Student Chapter May 2022 – May 2024

Delegate – Nuclear Engineering Student Delegation Sep. 2023, Sep. 2022

- Represented UIUC as a member of the Nuclear Engineering Student Delegation (NESD), a group of graduate and undergraduate nuclear engineering students from across the United States that meet in Washington, D.C. once a year to develop a policy statement and advocate for nuclear energy to their Congressional representatives
- Met with governmental and non-governmental organizations, such as the Department of Energy, the Office of Science Technology and Policy, the Nuclear Regulatory Commission, BWXT, and Clearpath to discuss current nuclear energy policy issues
- Policy statement available at: *nesd.org/policy-statements*

Outreach Chair – American Nuclear Society UIUC Student Chapter May 2021 – May 2022

- Organized monthly outreach events for the ANS Student Chapter at UIUC, including Boy Scouts troop outreach, Engineering Open House demonstration design and organization, and Nuclear Science Week outreach on the Main Quad activities

Student Conference Socials Chair – American Nuclear Society UIUC Student Chapter Nov. 2021 – April 2022

Social Chair – Women in Nuclear UIUC Student Chapter May 2021 – May 2022

Social Media Chair – Women in Nuclear UIUC Student Chapter Aug. 2020 – May 2021

Honors & Awards

University Nuclear Leadership Program Graduate Fellowship (Declined) April 2024
The U.S. Department of Energy, Office of Nuclear Energy

NSF Graduate Research Fellowship April 2024
The National Science Foundation

NPRE Outstanding Academic Achievement Award to a Graduating Senior Mar. 2024
Nuclear, Plasma, and Radiological Engineering Dept., UIUC

Kirkwood Women in Engineering Scholarship Sep. 2023
Grainger College of Engineering, UIUC

Nuclear Energy University Program Scholarship April 2023
The U.S. Department of Energy, Office of Nuclear Energy

Roy G. Post Waste Management Scholarship Dec. 2022
Roy G. Post Foundation

Nuclear Energy University Program Scholarship April 2022
The U.S. Department of Energy, Office of Nuclear Energy

Nuclear, Plasma, and Radiological Engineering Rising Undergraduate Research Award May 2021
Nuclear, Plasma, and Radiological Engineering Dept., UIUC

Skills

Languages: English and Working Spanish
Technical Skills: Python, PyTorch, TensorFlow, Git/GitHub, LaTeX, MATLAB
Machine Learning: Neural networks, interpretable/explainable AI, auto ML design

Research Areas: AI safety, model interpretability, AI for high-stakes applications